

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

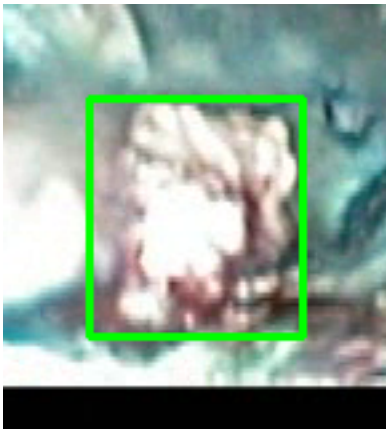
Video ID:	45267af3-9d21-47d7-bc68-08d0ad2a220b
Total Frames:	2974
Frame Rate:	25.0 fps (assumed)
Duration:	119.0 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	198	✓
RT-DETR Detections	1860	✓
MedSAM2 Detections	1831	✓
Consensus (3-Model Agreement)	15	✓
Risk Classification	Count	Percentage
HIGH RISK	0	0.0%
MEDIUM RISK	0	0.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	15	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — LIFTED_POLYP [LOW_RISK] (Tracked across 22 frames)

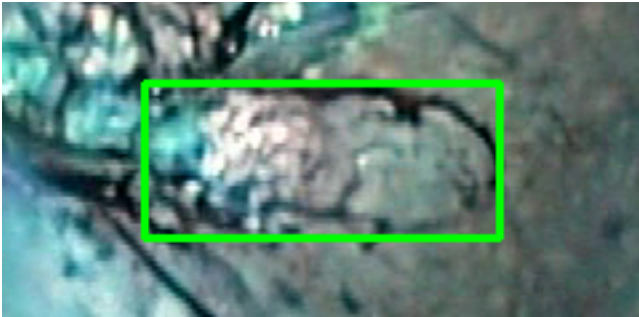


Feature	Value	Interpretation
Frame Index	287	Frame 287 of 2974
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 72.7%
Redness Score	0.000	Color-based vascularization
Relative Radius	6.5%	Polyp size as % of frame diagonal
Texture Score	0.460	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	67.4%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #2 — LIFTED_POLYP [LOW_RISK] (Tracked across 203 frames)

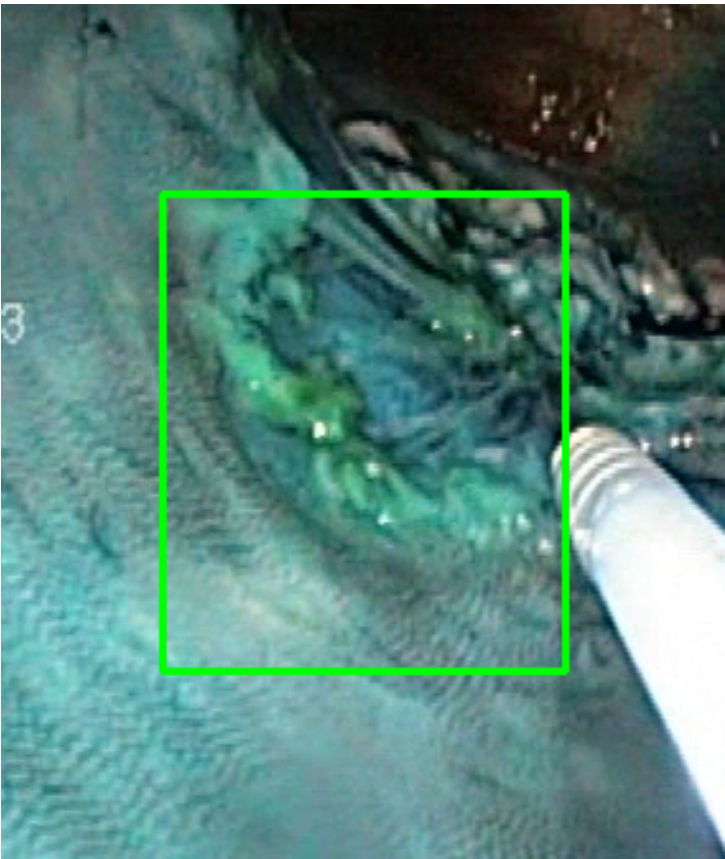


Feature	Value	Interpretation
Frame Index	2513	Frame 2513 of 2974
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 74.6%
Redness Score	0.002	Color-based vascularization
Relative Radius	15.1%	Polyp size as % of frame diagonal
Texture Score	0.517	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	69.4%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #3 — LIFTED_POLYP [LOW_RISK] (Tracked across 154 frames)



Feature	Value	Interpretation
Frame Index	526	Frame 526 of 2974
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 78.8%
Redness Score	0.000	Color-based vascularization
Relative Radius	15.5%	Polyp size as % of frame diagonal
Texture Score	0.429	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified

Detection Confidence	61.0%	YOLO / RT-DETR / track confidence
----------------------	-------	-----------------------------------

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #4 — LIFTED_POLYP [LOW_RISK] (Tracked across 20 frames)

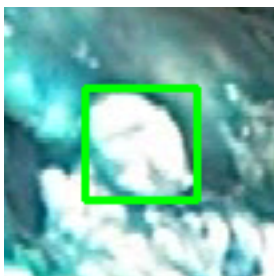


Feature	Value	Interpretation
Frame Index	982	Frame 982 of 2974
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 81.3%
Redness Score	0.000	Color-based vascularization
Relative Radius	5.2%	Polyp size as % of frame diagonal
Texture Score	0.472	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	55.9%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #5 — LIFTED_POLYP [LOW_RISK] (Tracked across 38 frames)



Feature	Value	Interpretation
Frame Index	1450	Frame 1450 of 2974
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 85.2%
Redness Score	0.000	Color-based vascularization
Relative Radius	8.9%	Polyp size as % of frame diagonal
Texture Score	0.589	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	57.5%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #6 — LIFTED_POLYP [LOW_RISK] (Tracked across 25 frames)



Feature	Value	Interpretation
Frame Index	1598	Frame 1598 of 2974
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 64.3%
Redness Score	0.001	Color-based vascularization
Relative Radius	14.5%	Polyp size as % of frame diagonal
Texture Score	0.204	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified

Detection Confidence	60.9%	YOLO / RT-DETR / track confidence
----------------------	-------	-----------------------------------

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #7 — LIFTED_POLYP [LOW_RISK] (Tracked across 24 frames)



Feature	Value	Interpretation
Frame Index	2597	Frame 2597 of 2974
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 76.3%
Redness Score	0.003	Color-based vascularization
Relative Radius	15.4%	Polyp size as % of frame diagonal
Texture Score	0.577	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	73.6%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #8 — LIFTED_POLYP [LOW_RISK] (Tracked across 24 frames)

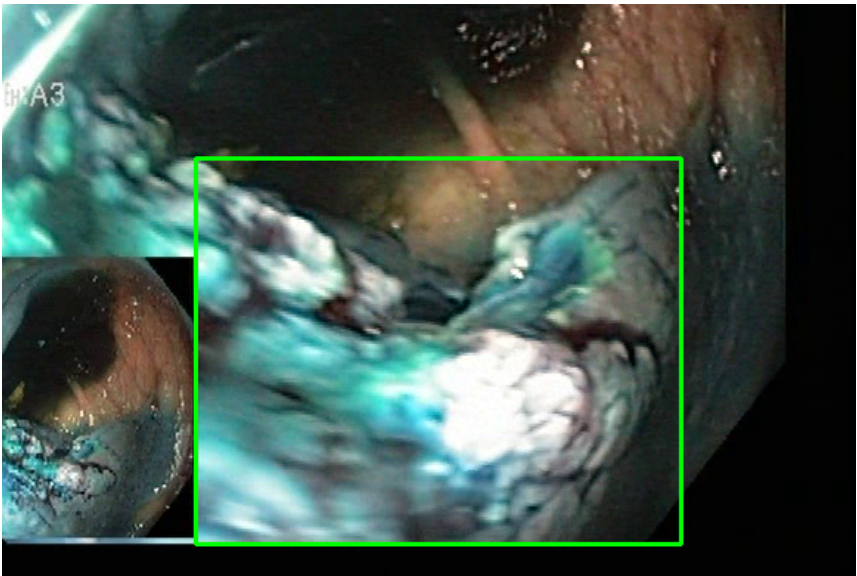


Feature	Value	Interpretation
Frame Index	1014	Frame 1014 of 2974
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 81.5%
Redness Score	0.000	Color-based vascularization
Relative Radius	4.3%	Polyp size as % of frame diagonal
Texture Score	0.522	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	55.7%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #9 — LIFTED_POLYP [LOW_RISK] (Tracked across 22 frames)



Feature	Value	Interpretation
Frame Index	2904	Frame 2904 of 2974
Detection Method	Detector Consensus (YOLO / RT-DETR)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 59.4%
Redness Score	0.010	Color-based vascularization
Relative Radius	22.5%	Polyp size as % of frame diagonal
Texture Score	0.134	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	66.8%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #10 — LIFTED_POLYP [LOW_RISK] (Tracked across 22 frames)



Feature	Value	Interpretation
Frame Index	287	Frame 287 of 2974
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 72.7%
Redness Score	0.000	Color-based vascularization
Relative Radius	6.5%	Polyp size as % of frame diagonal
Texture Score	0.460	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	66.0%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #11 — LIFTED_POLYP [LOW_RISK] (Tracked across 21 frames)



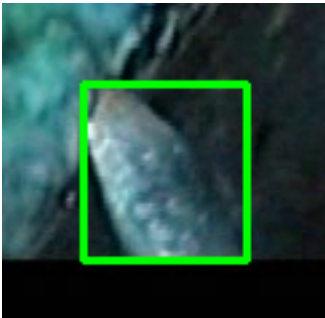
Feature	Value	Interpretation
Frame Index	1201	Frame 1201 of 2974
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 81.4%
Redness Score	0.000	Color-based vascularization
Relative Radius	10.8%	Polyp size as % of frame diagonal
Texture Score	0.355	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified

Detection Confidence	63.1%	YOLO / RT-DETR / track confidence
----------------------	-------	-----------------------------------

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #12 — LIFTED_POLYP [LOW_RISK] (Tracked across 21 frames)

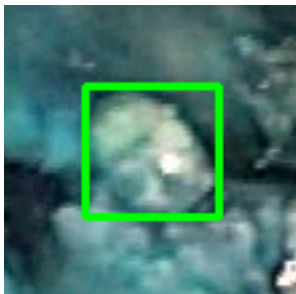


Feature	Value	Interpretation
Frame Index	1502	Frame 1502 of 2974
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 84.1%
Redness Score	0.001	Color-based vascularization
Relative Radius	8.4%	Polyp size as % of frame diagonal
Texture Score	0.527	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	58.0%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #13 — LIFTED_POLYP [LOW_RISK] (Tracked across 21 frames)

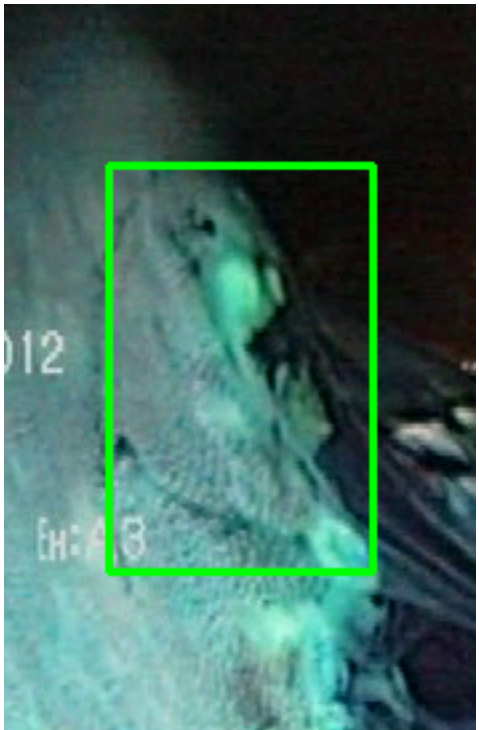


Feature	Value	Interpretation
Frame Index	1339	Frame 1339 of 2974
Detection Method	Detector Consensus (YOLO / RT-DETR)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 78.1%
Redness Score	0.000	Color-based vascularization
Relative Radius	12.1%	Polyp size as % of frame diagonal
Texture Score	0.500	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	49.6%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #14 — LIFTED_POLYP [LOW_RISK] (Tracked across 20 frames)

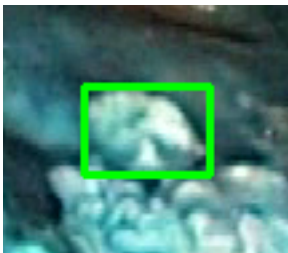


Feature	Value	Interpretation
Frame Index	623	Frame 623 of 2974
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Method Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 67.2%
Redness Score	0.004	Color-based vascularization
Relative Radius	17.2%	Polyp size as % of frame diagonal
Texture Score	0.158	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	67.1%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #15 — LIFTED_POLYP [LOW_RISK] (Tracked across 20 frames)



Feature	Value	Interpretation
Frame Index	982	Frame 982 of 2974
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 81.3%
Redness Score	0.000	Color-based vascularization
Relative Radius	5.2%	Polyp size as % of frame diagonal
Texture Score	0.472	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	63.9%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

CLINICAL IMPRESSION

Summary:

A total of 15 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.